Smoothness-Integrated Trend Distances with Nonzero Difference Drift

Mathematical formulation and numerical implications

Abstract

We compare time series using entire smoothness-indexed families of penalized least-squares (PLS) trends. Crucially, the penalized dth finite differences need not be centered at zero: we allow a nonzero reference drift μ in the objective $\|y-\tau\|_2^2+\lambda\|D_d\tau-\mu\mathbf{1}\|_2^2$. This leads to an affine trend estimator, a nonzero-drift smoothness limit, and a necessary distinction between the *penalty drift* μ and the *signed mean difference of two fitted trends* μ_Δ. Using the normalized coordinate $S\in[0,1]$ of the companion numerical study, we construct Hilbert-space global distances, time-local rolling distances, and smoothness–time dissimilarity surfaces. We give exact spectral representations with matching drift and a corrected formula for unequal drifts, plus a numerical example and monitoring cautions.

1 Penalized trend estimation with nonzero difference drift

Notation. Let $y=(y_1,\dots,y_N)^\top\in\mathbb{R}^N$ be an observed series at N aligned times; let d be a difference order with $1\le d<N$; let $D_d\in\mathbb{R}^{(N-d)\times N}$ represent the dth forward finite-difference operator; and let

$$Q=D_d^\top D_d\in\mathbb{R}^{N\times N},\qquad \mathbf{1}_r=(1,\dots,1)^\top\in\mathbb{R}^r. \tag{1}$$

The matrix Q is symmetric positive semidefinite and D_d has full row rank. Following the general nonzero-reference-difference formulation of Guerrero, introduce $\mu\in\mathbb{R}$, the target mean or reference level of the *dth differences*. Thus the penalty target is $\mu\mathbf{1}_{N-d}$, not a scalar subtracted from a vector. For $\lambda\ge0$, define

$$\widehat{\boldsymbol{\tau}}_{\lambda,\mu}(y)=\operatorname*{argmin}_{u\in\mathbb{R}^N}\left\{\|y-u\|_2^2+\lambda\|D_du-\mu\mathbf{1}_{N-d}\|_2^2\right\}. \tag{2}$$

The normal equations are

$$(I_N+\lambda Q)\widehat{\boldsymbol{\tau}}_{\lambda,\mu}=y+\lambda\mu D_d^\top\mathbf{1}_{N-d}. \tag{3}$$

Therefore, with the numerical manuscript's smoothing-matrix notation,

$$H_\lambda=(I_N+\lambda Q)^{-1},\qquad \widehat{\boldsymbol{\tau}}_{\lambda,\mu}(y)=H_\lambda\big(y+\lambda\mu D_d^\top\mathbf{1}_{N-d}\big). \tag{4}$$

The solution is unique because $I_N+\lambda Q$ is positive definite. At $\mu=0$ we recover the usual formula $H_\lambda y$ from the companion numerical paper; setting $\mu=0$ is an *assumption*, not a general identity.

The drift changes the limiting trend. Define the minimum-norm vector satisfying the reference-difference constraint,

$$\begin{aligned} b_\mu&=\mu D_d^\top(D_dD_d^\top)^{-1}\mathbf{1}_{N-d},\\ D_db_\mu&=\mu\mathbf{1}_{N-d},\quad b_\mu\perp\ker(D_d).\end{aligned} \tag{5}$$

Since $Qb_\mu=\mu D_d^\top\mathbf{1}_{N-d}$, equation (4) can be rearranged as

$$\widehat{\boldsymbol{\tau}}_{\lambda,\mu}(y)=b_\mu+H_\lambda(y-b_\mu). \tag{6}$$

Consequently,

$$\widehat{\boldsymbol{\tau}}_{0,\mu}(y)=y,\qquad \lim_{\lambda\to\infty}\widehat{\boldsymbol{\tau}}_{\lambda,\mu}(y)=P_{\ker(D_d)}y+b_\mu. \tag{7}$$

For $d=1$, a nonzero μ allows a linear limiting trend; for $d=2$, it allows a quadratic limiting trend. Thus, unlike the case $\mu=0$, strong smoothing need not make the dth difference vanish.

Known versus estimated drift. The drift μ may be specified in advance, shared across series, or estimated separately. A simple estimate from the observed dth differences, as in Guerrero, is

$$\widehat{\mu}(y)=\frac{1}{N-d}\mathbf{1}_{N-d}^\top D_dy. \tag{8}$$

If μ is estimated from y, the overall map from y to the trend is still linear for this choice, but its matrix is not generally H_λ; specifically,

$$\widehat{\boldsymbol{\tau}}_{\lambda,\widehat{\mu}(y)}(y)=\underbrace{H_\lambda\left[I_N+\frac{\lambda}{N-d}D_d^\top\mathbf{1}_{N-d}\mathbf{1}_{N-d}^\top D_d\right]}_{\widetilde{H}_\lambda}y. \tag{9}$$

The distinction affects spectral formulas and the interpretation of trace-based effective degrees of freedom.

2 The normalized smoothness coordinate S

We use the smoothness coordinate S from the companion manuscript. Because D_d has rank $N-d$, the matrix Q has d zero eigenvalues and $N-d$ positive eigenvalues $\rho_j>0$. Define

$$S(\lambda;N,d)=\frac{N}{N-d}\left(1-\frac{\operatorname{tr}(H_\lambda)}{N}\right)=1-\frac{1}{N-d}\sum_{j=1}^{N-d}\frac{1}{1+\lambda\rho_j}. \tag{10}$$

Then $S(0)=0$, $S(\infty)=1$, and $S'(\lambda)>0$ for finite $\lambda\ge0$. Thus its inverse $\lambda(S)$ exists for $0\le S<1$; $\lambda(1):=\infty$ is interpreted by a limit. Holding N, d, and D_d fixed gives a *common, data-independent calibration* across both series. For a supplied drift μ define the full trend path

$$\boldsymbol{\tau}_y(S;\mu):=\widehat{\boldsymbol{\tau}}_{\lambda(S),\mu}(y),\qquad S\in[0,1]. \tag{11}$$

When a distinct drift μ_i is used for each series, write $\tau_{y_i}(S) := \tau_{y_i}(S;\mu_i)$. The trace coordinate in (10) is defined using H_λ; for an estimated, data-dependent drift the effective linear map is $\widetilde{H}_\lambda$ in (9), so the trace of H_λ should be interpreted as the chosen *nominal Guerrero smoothness coordinate*, not automatically as the effective degrees of freedom of that adjusted estimator.

3 A global distance between entire trend families

The trend path is a vector-valued function of the smoothness coordinate. Work in the Hilbert space

$$\mathcal{H} = L^2([0,1];\mathbb{R}^N), \qquad \langle f, g\rangle_{\mathcal{H}} = \frac{1}{N}\int_0^1 f(S)^\top g(S)\, dS. \tag{12}$$

Since $\|H_\lambda\|_{2\to 2} \le 1$ and b_μ is fixed, (6) shows that each trend path is bounded and therefore belongs to $\mathcal{H}$. Given series $y_1, y_2 \in \mathbb{R}^N$, define

$$\mathcal{D}_{\mathrm{PLS}}(y_1, y_2) = \left[\frac{1}{N}\int_0^1 \|\tau_{y_1}(S) - \tau_{y_2}(S)\|_2^2\, dS\right]^{1/2}. \tag{13}$$

This measures disagreement over *all* smoothing levels. The pointwise-in-smoothness RMS distance is

$$d(S) = \frac{1}{\sqrt{N}}\|\tau_{y_1}(S) - \tau_{y_2}(S)\|_2, \qquad \mathcal{D}_{\mathrm{PLS}}^2 = \int_0^1 d(S)^2\, dS. \tag{14}$$

The Hilbert-space norm is a metric on trend paths (modulo equality almost everywhere). Its pullback to observed series is a pseudometric, and is a metric whenever the chosen smoothing-and-drift mapping is injective almost everywhere. In particular, if the same fixed μ and the same calibration are used, the result is a metric on $\mathbb{R}^N$.

4 A nonzero *mean difference* between fitted trends

Here the symbol μ_Δ is intentionally *different* from the penalty-drift parameter μ in Section 1. Let $\mathbf{1}_N$ be the constant vector and set

$$\Delta(S) = \tau_{y_2}(S) - \tau_{y_1}(S), \qquad \mu_\Delta(S) = \frac{1}{N}\mathbf{1}_N^\top \Delta(S). \tag{15}$$

No zero-mean assumption is made. Split into an average level and centered discrepancy,

$$\Delta(S) = \mu_\Delta(S)\mathbf{1}_N + \widetilde{\Delta}(S), \qquad \mathbf{1}_N^\top \widetilde{\Delta}(S) = 0. \tag{16}$$

Orthogonality gives the global decomposition

$$\mathcal{D}_{\mathrm{PLS}}^2 = \underbrace{\int_0^1 \mu_\Delta(S)^2\, dS}_{\mathcal{D}_{\mathrm{level}}^2} + \underbrace{\frac{1}{N}\int_0^1 \|\widetilde{\Delta}(S)\|_2^2\, dS}_{\mathcal{D}_{\mathrm{shape}}^2}. \tag{17}$$

The scale-averaged *signed* level gap is

$$\overline{\mu}_\Delta = \int_0^1 \mu_\Delta(S)\, dS. \tag{18}$$

This signed summary preserves direction; the nonnegative level component $\mathcal{D}_{\mathrm{level}}$ does not.

Why nonzero penalty drift does not destroy mean preservation. For every difference order $d \ge 1$, $D_d\mathbf{1}_N = 0$. Hence $H_\lambda\mathbf{1}_N = \mathbf{1}_N$ and $\mathbf{1}_N^\top H_\lambda D_d^\top = 0$, regardless of μ. Thus the affine estimator in (4) satisfies

$$\frac{1}{N}\mathbf{1}_N^\top \tau_y(S;\mu) = \overline{y} \qquad \text{for all } S \in [0,1] \text{ and all } \mu. \tag{19}$$

Consequently, even for *different, nonzero* drifts μ_1 and μ_2,

$$\mu_\Delta(S) = \overline{y}_2 - \overline{y}_1 =: \mu_0, \qquad \mathcal{D}_{\mathrm{PLS}}^2 = \mu_0^2 + \mathcal{D}_{\mathrm{shape}}^2. \tag{20}$$

Here μ_0 is an observed-level mean difference; it is *not* the penalty drift μ. These identities require the ordinary unweighted least-squares data term and a difference operator annihilating constants.

5 Numerical integration in the normalized coordinate

For a grid $0 = S_0 < S_1 < \cdots < S_M = 1$, calculate both trend paths, including their chosen reference drifts, and apply trapezoidal quadrature:

$$\mathcal{D}_{\mathrm{PLS}}(y_1, y_2) \approx \left\{\sum_{j=0}^{M-1} \frac{S_{j+1} - S_j}{2}\left[d(S_j)^2 + d(S_{j+1})^2\right]\right\}^{1/2}. \tag{21}$$

A first grid is $S_j = j/10$; refine and check numerical stability. The endpoint $S = 1$ must be evaluated from (7), not by blindly replacing $\lambda = \infty$ with a large finite number.

6 Spectral form: common and unequal drifts

Assume the same N, d, D_d and hence the same $\lambda(S)$ for both series. Let $\Delta y = y_2 - y_1$, let $b_i = b_{\mu_i}$ from (5), and let $\Delta b = b_2 - b_1$. Equation (6) gives

$$\Delta(S) = H_{\lambda(S)}(\Delta y - \Delta b) + \Delta b. \tag{22}$$

Write $Q = U\,\mathrm{diag}(\rho_1, \ldots, \rho_N)U^\top$, with $U^\top U = I$, and let u_j denote column j of U. The general formula is

$$\mathcal{D}_{\mathrm{PLS}}^2 = \frac{1}{N}\sum_{j=1}^N \int_0^1 \left[\frac{u_j^\top \Delta y}{1 + \lambda(S)\rho_j} + \left(1 - \frac{1}{1 + \lambda(S)\rho_j}\right) u_j^\top \Delta b\right]^2 dS. \tag{23}$$

In particular, if the two reference drifts are equal ($\mu_1 = \mu_2$, not necessarily zero), then $\Delta b = 0$ and

$$\mathcal{D}_{\mathrm{PLS}}^2 = \frac{1}{N}\Delta y^\top M \Delta y, \qquad M = \int_0^1 H_{\lambda(S)}^2\, dS, \tag{24}$$

with the simpler spectral weights

$$\mathcal{D}_{\mathrm{PLS}}^2 = \frac{1}{N}\sum_{j=1}^N \left[\int_0^1 \frac{dS}{(1 + \lambda(S)\rho_j)^2}\right](u_j^\top \Delta y)^2. \tag{25}$$

A nonzero *common* μ cancels from the difference and leaves this spectral metric unchanged. If μ_i are estimated separately

from the data, Δb generally does not vanish; use (23), not (25). The nominal coordinate S is still common when N, d, D_d are fixed. If one instead calibrates smoothness separately in a data-dependent way, the single-matrix formulas require further modification.

7 From global similarity to time-localized differences

Let $t \in \{1, \dots, N\}$ index aligned observation times and set

$$\begin{aligned} \tau_i(S,t) &= [\boldsymbol{\tau}_{y_i}(S)]_t, \\ \Delta(S,t) &= \tau_2(S,t) - \tau_1(S,t), \\ \delta(S,t) &= |\Delta(S,t)|. \end{aligned} \tag{26}$$

The signed difference Δ indicates *which* series has a higher estimated trend, whereas δ measures the magnitude of the difference at one time and one smoothing level. Neither quantity alone is a rolling-window distance.

7.1 A worked example at one smoothing level

To illustrate the distinction, fix a smoothness level S and suppose two seven-point trend vectors are

$$\begin{aligned} \boldsymbol{\tau}_{y_1}(S) &= (1,1,1,1,1,1,1)^\top, \\ \boldsymbol{\tau}_{y_2}(S) &= (1,1,1,1,3,5,7)^\top. \end{aligned} \tag{27}$$

These are *illustrative vectors*, not claimed to be outputs of a particular PLS estimator. Their absolute difference is

$$\big(\delta(S,1), \dots, \delta(S,7)\big) = (0,0,0,0,2,4,6). \tag{28}$$

The series coincide for times 1–4 and start differing at time 5.

7.2 Scale-specific rolling distance

Choose a window length $w \in \{1, \dots, N\}$. For $t = w, \dots, N$, define

$$D_w(S,t) = \left[\frac{1}{w} \sum_{k=t-w+1}^{t} \Delta(S,k)^2 \right]^{1/2}. \tag{29}$$

The map $(S,t) \mapsto D_w(S,t)$ is a *scale–time dissimilarity surface*: for each smoothness level it measures the root-mean-square difference over the most recent w observations. A heatmap of this surface can distinguish differences that are visible only at some smoothness levels from those that persist over a broad range of scales.

For the vectors in (27), $w = 3$ gives (the signed mean $\mu_3(S,t)$ is defined in Section 7.3) The pointwise difference responds immediately at time 5, while the rolling RMS accumulates the discrepancy over the selected window.

7.3 Rolling signed mean and mean-adjusted variation

To distinguish a persistent directional shift from fluctuating mismatches, retain the *signed* window mean

$$\mu_w(S,t) = \frac{1}{w} \sum_{k=t-w+1}^{t} \Delta(S,k). \tag{30}$$

Its sign tells us which trend tends to be higher in that window (recall $\Delta = \tau_2 - \tau_1$). Define the within-window centered mean-square difference using divisor w,

$$v_w(S,t) = \frac{1}{w} \sum_{k=t-w+1}^{t} \big[\Delta(S,k) - \mu_w(S,t)\big]^2. \tag{31}$$

Expanding the square and using the definition of μ_w proves

$$D_w(S,t)^2 = \mu_w(S,t)^2 + v_w(S,t). \tag{32}$$

This decomposition explains *why* the rolling RMS can be large: either the signed difference has a substantial nonzero mean, or the differences fluctuate around their mean (or both). It holds even when the mean is not zero.

7.4 Rolling distance integrated across all smoothness levels

Rather than selecting a single S, integrate the squared rolling discrepancy over $S \in [0,1]$:

$$\begin{aligned} \mathcal{D}_w(t)^2 &= \int_0^1 D_w(S,t)^2 \, dS \\ &= \frac{1}{w} \sum_{k=t-w+1}^{t} \int_0^1 \Delta(S,k)^2 \, dS. \end{aligned} \tag{33}$$

For each fixed (w,t), the restricted paths $S \mapsto (\tau_i(S, t-w+1), \dots, \tau_i(S,t))$ are elements of $L^2([0,1]; \mathbb{R}^w)$. Equation (33) measures their Hilbert-space distance with a factor $1/w$ in the inner product. It can be a pseudometric in the original series because restricting attention to a window may discard distinguishing information. For the full window, $w = N$ and $t = N$, the rolling-integrated distance reduces exactly to the global distance in (13).

Integrating over all smoothing levels leads to three complementary summaries:

$$L_w(t)^2 = \int_0^1 \mu_w(S,t)^2 \, dS, \quad \text{rolling level component,} \tag{34}$$

$$V_w(t)^2 = \int_0^1 v_w(S,t) \, dS, \quad \text{rolling centered component,} \tag{35}$$

$$\overline{\mu}_w(t) = \int_0^1 \mu_w(S,t) \, dS, \quad \text{signed, scale-averaged shift.} \tag{36}$$

Time t	Pointwise $\delta(S,t)$	Signed $\mu_3(S,t)$	Rolling $D_3(S,t)$
3	0	0	0
4	0	0	0
5	2	2/3	$\sqrt{4/3} \approx 1.155$
6	4	2	$\sqrt{20/3} \approx 2.582$
7	6	4	$\sqrt{56/3} \approx 4.320$

Table 1. Pointwise discrepancy, signed rolling mean, and three-point rolling RMS for the synthetic example.

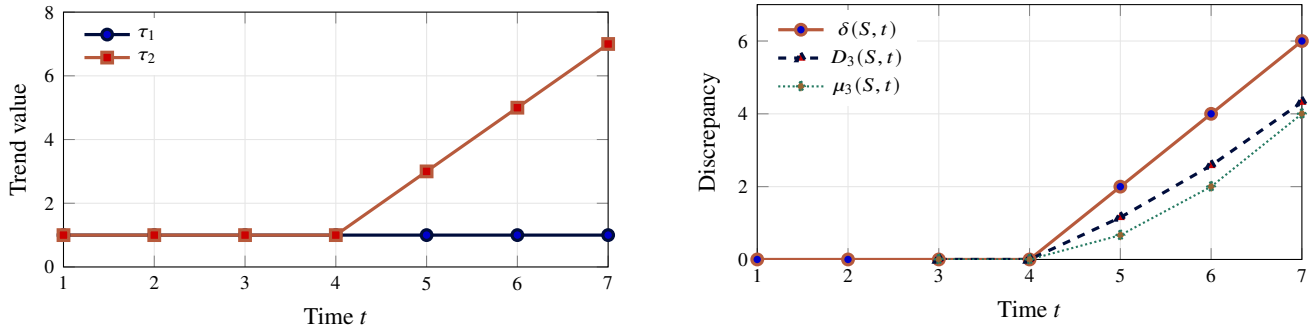


Figure 1. Synthetic illustration at a fixed smoothness level. Left: two trend vectors. Right: the pointwise absolute difference and the three-observation rolling RMS and signed rolling mean.

By (32),

$$\mathcal{D}_w(t)^2 = L_w(t)^2 + V_w(t)^2. \tag{37}$$

Notice that $L_w(t) \geq |\overline{\mu}_w(t)|$ by Cauchy–Schwarz; averaging signed differences across scales can hide opposing shifts. For the example in (27), at $t = 7$ and $w = 3$, the window differences are $(2, 4, 6)$, giving

$$\begin{aligned} \mu_3(S,7) &= 4, \qquad v_3(S,7) = \frac{8}{3}, \\ D_3(S,7)^2 &= 16 + \frac{8}{3} = \frac{56}{3}. \end{aligned} \tag{38}$$

The example is at a single selected smoothness level; it does not specify a full S-indexed trend family.

A temporal change statistic can be defined for $t \geq w + 1$ as

$$C_w(t) = \mathcal{D}_w(t) - \mathcal{D}_w(t-1). \tag{39}$$

Positive values signal an increase in recent dissimilarity, but $C_w(t)$ is not by itself a calibrated statistical change-point test. The signed rolling mean $\overline{\mu}_w(t)$ from (36) summarizes the direction of the difference, whereas $\mathcal{D}_w(t)$ measures its size; retain both.

8 Computation and operator structure of rolling comparisons

On the smoothness grid from Section 5, compute $D_w(S_j,t)^2$ at each grid point and apply the same quadrature rule:

$$\mathcal{D}_w(t)^2 \approx \sum_{j=0}^{M-1} \frac{S_{j+1} - S_j}{2} \left[D_w(S_j,t)^2 + D_w(S_{j+1},t)^2 \right]. \tag{40}$$

For efficient evaluation, compute a time-indexed integrated squared discrepancy,

$$g(k) = \int_0^1 \Delta(S,k)^2 \, dS, \qquad \mathcal{D}_w(t)^2 = \frac{1}{w} \sum_{k=t-w+1}^{t} g(k). \tag{41}$$

Thus $\mathcal{D}_w(t)^2$ is a moving average of $g(k)$. In parallel, for each sampled S maintain rolling sums of $\Delta(S,k)$ and $\Delta(S,k)^2$. Their ratio to w yields $\mu_w(S,t)$ and $D_w(S,t)^2$, while $v_w(S,t) = D_w(S,t)^2 - \mu_w(S,t)^2$ (clipped at zero to handle small floating-point roundoff). Apply smoothness quadrature to compute L_w, V_w, and $\overline{\mu}_w$. Once the sampled trends and per-time integrals are computed, all rolling distances for a fixed w can be obtained in $O(N)$ time using cumulative sums, rather than recomputing each window from scratch. The full (S,t) surface can likewise be computed using rolling sums separately at each sampled smoothness level.

Under the *common-calibration and common-drift assumptions* of Section 6, let $P_{w,t} \in \mathbb{R}^{w \times N}$ select the coordinates $t - w + 1, \ldots, t$. Then

$$\begin{aligned} \mathcal{D}_w(t)^2 &= \frac{1}{w} \int_0^1 \left\| P_{w,t} H_{\lambda(S)} \Delta y \right\|_2^2 \, dS \\ &= \Delta y^\top M_{w,t} \Delta y, \\ M_{w,t} &= \frac{1}{w} \int_0^1 H_{\lambda(S)} P_{w,t}^\top P_{w,t} H_{\lambda(S)} \, dS. \end{aligned} \tag{42}$$

The matrix $M_{w,t}$ is positive semidefinite. Unlike the global metric matrix M in (24), it need not share the eigenvectors of L, since $P_{w,t}^\top P_{w,t}$ need not commute with L. If the two series use different drifts or different calibrations, the operator representation must incorporate those differences; nevertheless, (33) remains well-defined.

9 Application to agricultural monitoring and study design

Suppose two fields have comparable historical trajectories of an agricultural index such as NDMI. The functions

$$t \longmapsto \mathcal{D}_w(t), \qquad t \longmapsto \overline{\mu}_w(t), \qquad (S,t) \longmapsto D_w(S,t) \tag{43}$$

answer complementary questions: *when* do the trends diverge, *in which direction*, and *at which smoothing scales* are discrepancies visible? A persistent nonzero signed mean can indicate an offset between fields, while large centered variation can indicate different temporal shapes. A persistent difference over many scales may warrant investigation of water stress, precipitation, crop cover, management, or observation quality. However, divergence of a vegetation index is not, by itself, proof of any particular cause.

Statistical interpretation of $\mu \neq 0$. An observed $\mu_w(S,t) \neq 0$ is a descriptive statement, not necessarily evidence of a statistically significant systematic shift. One could test $H_0 : \mathbb{E}[\Delta(S,k)] = 0$ against a nonzero-mean alternative for a defined population and time regime, but differences across time and smoothing levels are dependent. Any uncertainty quantification must account for temporal correlation, overlapping windows, multiple smoothness comparisons, and measurement error; block-bootstrap or other dependence-aware methods are candidates.

Retrospective versus online monitoring. A full-sample PLS trend at time k generally uses observations after k. Hence the retrospective $\mathcal{D}_w(t)$ in (33) may show divergence before information would have been available in real time. An online variant must recompute trend estimates from data available *by time t only*. If $\tau_i^{[t]}(S,k)$ denotes such an estimate, set

$$\left[\mathcal{D}_w^{\text{online}}(t)\right]^2 = \frac{1}{w} \sum_{k=t-w+1}^{t} \int_0^1 \left|\tau_1^{[t]}(S,k) - \tau_2^{[t]}(S,k)\right|^2 dS. \tag{44}$$

Because past estimates can be revised when new observations arrive, changes in (44) need to be interpreted under a stated online estimation protocol. The smoothness calibration itself must also use only past and present information.

Suggested validation. Compare global, rolling, and scale–time distances with raw-series rolling RMS, single-scale rolling RMS, and established change-detection baselines. Vary w, the smoothness-grid resolution, and the calibration rule. Assess false alarms and detection delays in simulations with known onset times, then investigate real field pairs with independent agronomic evidence.

10 Interpretation and extensions

Equation (13) compares *levels and shapes* across all smoothness scales. If shape alone is of interest, a stated centering or normalization should be applied to X_1 and X_2 before trend fitting; such preprocessing may change the metric properties on raw series. A smoothness-weighted variant (distinct from the rolling window length w) is

$$\mathcal{D}_\omega(y_1,y_2)^2 = \frac{1}{N}\int_0^1 \omega(S)\,\|\mathbf{\Delta}(S)\|_2^2\,dS, \qquad \omega(S) \geq 0, \qquad \int_0^1 \omega(S)\,dS = 1. \tag{45}$$

The choice of ω expresses which smoothness levels matter most.

11 Questions for validation

1. Is smoothness S calibrated in the same way across series, or through series-specific maps $\lambda_y(S)$?
2. Does the integrated discrepancy add meaningful information over raw Euclidean distance and a distance computed at one smoothing level?
3. Is the result stable under grid refinement, measurement noise, different time-series lengths, and alternative roughness operators?
4. Should comparisons be invariant to additive offsets or multiplicative scales? Should nonzero mean difference be reported separately from the centered shape discrepancy, and which normalization is justified?
5. Which window lengths w balance detection delay against sensitivity to measurement noise? At which smoothness levels does a divergence first become detectable?
6. Can an entirely causal trend-estimation and smoothness-calibration protocol avoid future-data leakage when reporting detection times?

12 Notation and source conventions

The primary notation follows *Numerical Selection of Forecast-Optimal Smoothness in Penalized Trend Estimation*: the observed vector is y, the finite-difference matrix is D_d, the roughness matrix is $Q = D_d^\top D_d$, the smoothing matrix is H_λ, and the normalized smoothness coordinate is $S \in [0,1]$. The generalized nonzero-drift penalty follows Guerrero's treatment of the reference level for finite differences. Two different meanings of "mean" must not be conflated: μ is the scalar target for the *dth differences*, whereas μ_Δ is the average signed difference between the *fitted trend levels*.

References

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